COMPUTER FORENSICS CASE STUDY REPORT

SBT-DF204 — Case Study 2

Reconstructing Suspected Drug Trafficking From Chrome Web History

Trainee Name: FASINU TEMIDAYO LUCKY
Registration Number: 2025/FWSD/11378

Course: SBT-DF204 Computer Forensics Case Studies

Case Title: Reconstructing Suspected Drug Trafficking From Chrome Web History

Evidence Item: Chrome History SQLite database ("History")

Analysis Workstation: Kali Linux VM
Report Date: 12 October 2026

1. Case Reference and Student Identification

Field	Value
Full name	Fasinu Temidayo Lucky
Registration number	2025/FWSD/11378
Course code	SBT-DF204 — Computer Forensics Case Studies
Assessment	Case Study 2 — Individual browser history investigation
Case title	Reconstructing Suspected Drug Trafficking From Chrome Web History
Evidence item	Google Chrome History SQLite database, file name "History"
Analysis workstation	Kali Linux VM
Examination tool	sqlite3 (command-line SQLite shell)

This report is submitted individually. The SQL queries, interpretation, wording, screenshots, timeline and evidence log that follow are my own work, produced by direct forensic examination of the supplied Chrome History database. The case requires interpretation of the supplied database only; no live website, account, message, exchange account or cryptocurrency wallet referenced in the history was visited or interacted with during this investigation.

2. Executive Conclusion

Examination of the Chrome History database recovered from the subject device shows 74 browsing visits across 57 distinct URLs, spanning 2022-04-19 13:54:18 UTC to 15:02:56 UTC (a session window of 4,118 seconds). Within that window, the browser profile was used to: search for and log into Craigslist; complete a seven-step posting workflow that published an advertisement titled "cheaper than Rx supplements" under posting ID 7473121658; access a Gmail inbox (unsub.fscs@gmail.com) containing an "order" thread and a thread referencing a Bitcoin transaction ID (517b2156914944339a96137ad8978408ea52b2fc144c98d3b0b16b21888afdc5); follow a Gmail-sourced redirect to an Imgur image (https://imgur.com/dTgrkP7) that corresponds to a locally downloaded file, proof_of_payment.png; look up the same transaction ID on blockchain.com and mempool.space; and sign in to the Kraken cryptocurrency exchange, where a funding-overview page and a ledger-history page were viewed showing two USD balance figures.

The pattern and sequence of these artefacts — an advertisement for an ambiguously described product, buyer contact through the same mailbox, a

blockchain transaction ID cross-verified on two independent explorers, and retained proof of payment — is consistent with the case hypothesis that the browser was used to advertise a controlled item, communicate with a buyer, and verify receipt of cryptocurrency payment. This conclusion is supported directly by browser-history artefacts (URLs, titles, timestamps, transitions, and one downloaded file) and is stated with MEDIUM confidence as an activity pattern. It is explicitly NOT a conclusive finding that any named individual personally performed these actions: a Chrome browser profile identifies a profile and a device, not conclusively the human operator at the keyboard at every moment, and the history contains no content capture, account-holder identity document, or device-user log that would close that gap. Section 7 sets out the full attribution assessment, including at least three specific limitations, as required by the assignment brief.

3. Evidence Acquisition and Integrity

(Investigation Question 1)

Investigation question: What evidence file did you acquire and how did you preserve it?

Field	Value
Evidence file name	History
Source	https://raw.githubusercontent.com/frankwxu/digital-forensics-lab/main/Digital_Currency/LabFiles/History
Download date/time	2026-10-12 05:19:27 – 05:19:28
File size	196,608 bytes (192 KB)
File type	SQLite 3.x database, last written using SQLite version 3036000, 41 database pages, schema 4, UTF-8
SHA-256 (computed)	b991b0fa69662c44caf599e468c8088576ebd9cebbf46d02c89cf51bcde3fd19
SHA-1 (computed, secondary check)	e14ad4c53c44cafd286c78bd4073e1fd02d1af00
MD5 (computed, secondary check)	0c594e9de87348c089554953a67e2c74

Field	Value
Working copy	working/History_working — created with cp -v from evidence/History; SHA-256 verified byte-identical to the original
Original file handling	Set to read-only (chmod 444) immediately after hashing and before any examination; all SQL analysis performed on the working copy only

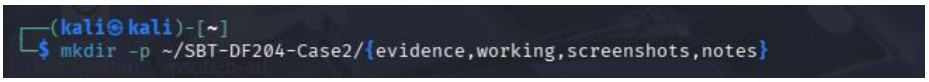


Figure 3.1 — Evidence folder structure created before acquisition: evidence, working, screenshots, notes

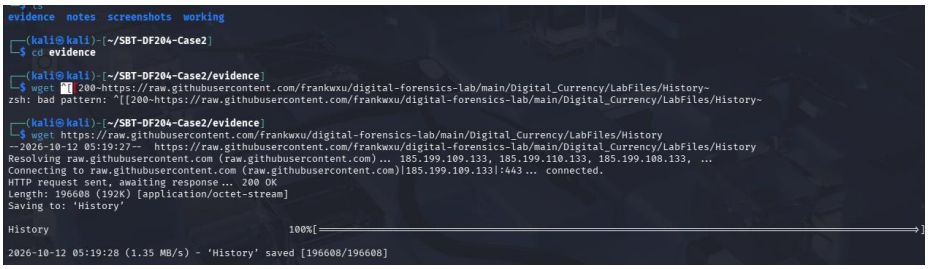


Figure 3.2 — Acquiring the History file with wget into the evidence folder (196,608 bytes, HTTP 200 OK)

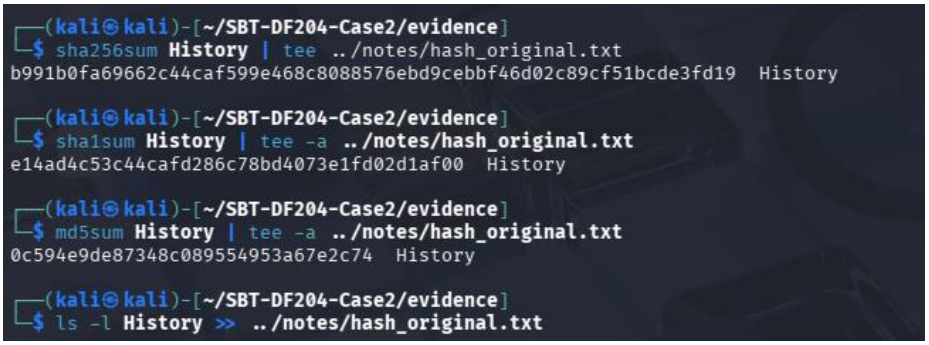


Figure 3.3 — SHA-256, SHA-1 and MD5 of the original History file, recorded to notes/hash_original.txt



Figure 3.4 — Working copy created with cp -v: evidence/History → working/History_working

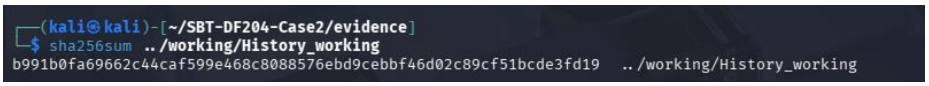


Figure 3.5 — SHA-256 of the working copy, confirmed identical to the original (b991b0fa…)

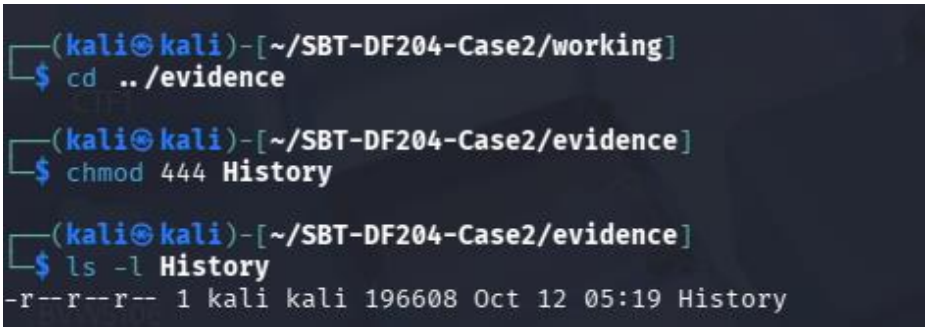


Figure 3.6 — Original evidence file set to read-only (chmod 444); ls -l confirms -r--r--r--



Figure 3.7 — File type identification via the file command: SQLite 3.x database, schema 4, UTF-8

The SHA-256 value computed for the working copy (b991b0fa69662c44caf599e468c8088576ebd9cebbf46d02c89cf51bcde3fd19) matches the SHA-256 computed for the original evidence file exactly, confirming the working copy is byte-identical and that no modification occurred during the copy operation. All subsequent SQL queries in this report were executed exclusively against working/History_working; the original evidence/History file was never opened for examination after hashing, and was protected read-only throughout. This chain of custody allows another examiner to independently verify they are analysing the same evidence and to reproduce every query and finding in this report.

Evidence reference: E01.

4. Database Method (Investigation Question 2)

Investigation question: What browser-history tables and fields are relevant?

The database schema was enumerated before any content query was run, since Chrome's schema can differ by version. The sqlite_master query returned 17 tables; four were directly relevant to this case (urls, visits, downloads, downloads_url_chains), and a fifth (keyword_search_terms) supplied corroborating evidence of the user's search activity.



Figure 4.1 — sqlite3 History_working "SELECT name FROM sqlite_master WHERE type='table' ORDER BY name;" — 17 tables listed

Table	Row count	Role in this investigation
urls	57	One row per distinct URL visited; holds url, title, visit_count, typed_count
visits	74	One row per navigation event; holds visit_time (WebKit timestamp), from_visit, transition, visit_duration — the backbone of the reconstructed timeline
downloads	1	The single retained file, proof_of_payment.png, with its source tab_url, byte counts and timestamps
downloads_url_chains	1	The URL chain associated with the one download record
keyword_search_terms	4	Omnibox search terms typed by the user: craigslist, gmail, blockchain explorer, kraken



Figure 4.2 — PRAGMA table_info(urls) and PRAGMA table_info(visits): column definitions for the two core tables

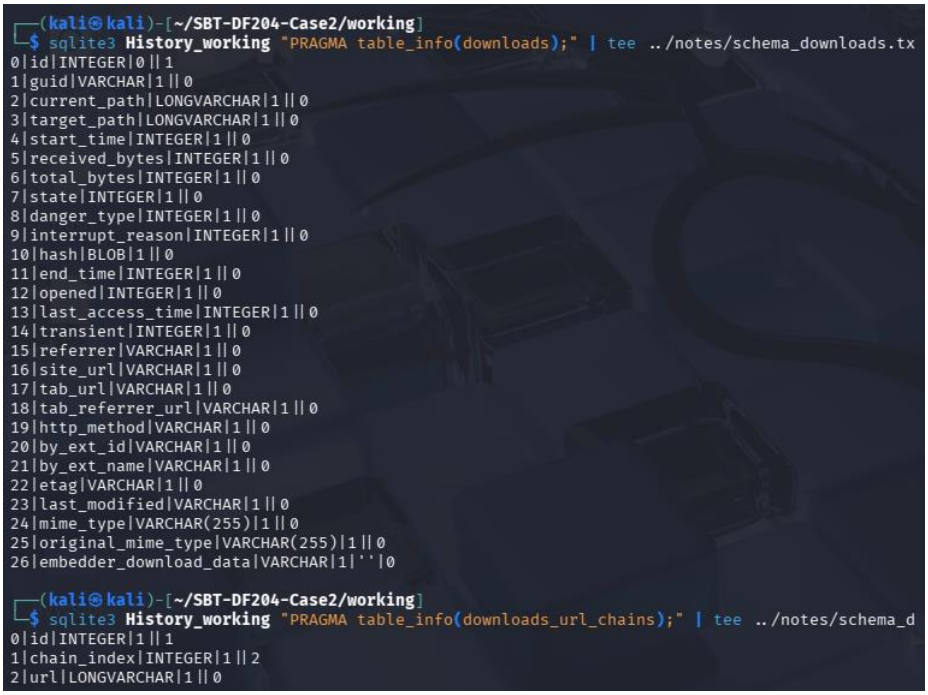


Figure 4.3 — PRAGMA table_info(downloads) and PRAGMA table_info(downloads_url_chains): column definitions

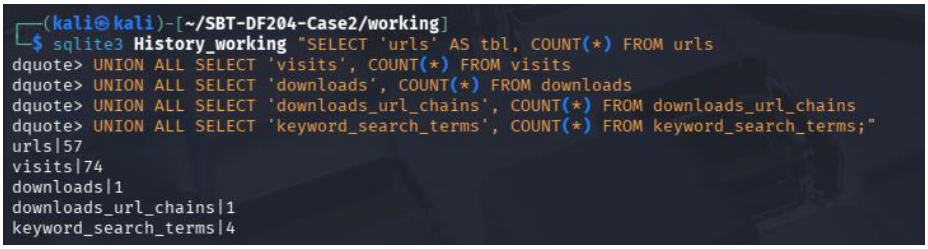


Figure 4.4 — UNION ALL row-count query confirming urls=57, visits=74, downloads=1, downloads_url_chains=1, keyword_search_terms=4

4.1 Relevant fields

Table.Field	Meaning / use in this report
urls.url / urls.title	The destination address and page title of each visited resource
urls.visit_count / urls.typed_count	How many times a URL was visited / how many of those visits began with typed input
visits.visit_time	WebKit/Chrome timestamp — microseconds since 1601-01-01 00:00:00 UTC — converted to UTC for the timeline
visits.from_visit	The visit_id of the referring navigation; 0 marks the start of a new navigation chain
visits.transition	A bitmask: low byte (AND 0xFF) gives the base transition type (link, typed, omnibox, form submit, etc.); high bits are qualifier flags (chain start, client/server redirect, forward/back)
visits.visit_duration	Time (microseconds) the page remained the active tab before the next navigation

Table.Field	Meaning / use in this report
downloads.current_path / target_path / tab_url	The saved file's path and the page it was downloaded from
downloads.received_bytes / total_bytes / mime_type	Download size and content type, used to confirm the file is a genuine, complete image
downloads_url_chains.url	The download's source URL chain, cross-referenced against the urls/visits records for the same page

Investigation question: When did the recorded web activity occur? The WebKit-to-UTC conversion method used throughout this report is set out here.

```
SELECT v.id AS visit_id, u.id AS url_id, u.url, u.title,
  datetime(v.visit_time/1000000 + strftime('%s','1601-01-01'), 'unixepoch') AS visit_utc,
  v.from_visit, v.transition, v.visit_duration/1000.0 AS duration_ms
FROM visits AS v JOIN urls AS u ON v.url = u.id
ORDER BY v.visit_time;
```

```
┌──(kali㉿kali)-[~/SBT-DF204-Case2/working]
└─$ sqlite3 History_working -header -column "SELECT id, url, visit_time, from_visit, transitio
n, visit_duration FROM visits ORDER BY id LIMIT 15;"
id  url  visit_time         from_visit  transition   visit_duration
--  ---  -----------------  ----------  -----------  --------------
1   1    13294850058231547  0           805306373    0
2   1    13294850059218619  0           805306368    727413
3   2    13294850059946032  2           268435456    0
4   3    13294850059946032  3           -2147483648  0
5   4    13294850059946032  4           -1610612736  2465460
6   5    13294850062411492  5           268435456    0
7   6    13294850062411492  6           -1610612736  17456657
8   5    13294850079868149  7           805306375    23439248
9   2    13294850103307397  8           268435456    0
10  4    13294850103307397  9           -1610612736  2664177
11  7    13294850105971574  10          268435456    0
12  8    13294850105971574  11          -2147483648  0
13  9    13294850105971574  12          -1610612736  4273081
14  10   13294850110244655  13          805306375    5599315
15  11   13294850115843970  14          805306375    177674775
```

Figure 4.5 — First 15 rows of the raw visits table (id, url, visit_time, from_visit, transition, visit_duration) before conversion

```
┌──(kali㉿kali)-[~/SBT-DF204-Case2/working]
└─$ sqlite3 History_working -header -column "SELECT id, substr(url,1,80) AS url_short, substr(
title,1,60) AS title_short, visit_count FROM urls ORDER BY id LIMIT 20;"
id  url_short                                                                        title_sh
ort                                                                                visit_count
--  ---------------------------------------------------------------------------------  --------
--  ------------------------------------------------------------  -----------
1   https://www.google.com/search?q=craigslist&rlz=1C1ONGR_enUS989US989&oq=craigslis  craigsli
st - Google Search                                                                 2
2   https://www.craigslist.org/                                                    craigsli
st: baltimore jobs, apartments, for sale, services,                                2
3   https://geo.craigslist.org/                                                    craigsli
st: baltimore jobs, apartments, for sale, services,                                1
4   https://baltimore.craigslist.org/                                              craigsli
st: baltimore jobs, apartments, for sale, services,                                2
5   https://accounts.craigslist.org/login/home                                     craigsli
st account                                                                         2
6   https://accounts.craigslist.org/login?rp=%2Flogin%2Fhome&rt=L                  craigsli
st - account log in                                                                1
7   https://post.craigslist.org/c/bal                                              baltimor
e | choose type                                                                    1
8   https://post.craigslist.org/k/sHbOUOi_7BG68qUDIbeTbQ/bQTi5                     baltimor
e | posting confirmation                                                           2
9   https://post.craigslist.org/k/sHbOUOi_7BG68qUDIbeTbQ/bQTi5?s=type              baltimor
e | choose type                                                                    1
10  https://post.craigslist.org/k/sHbOUOi_7BG68qUDIbeTbQ/bQTi5?s=cat               baltimor
e | choose category                                                                1
11  https://post.craigslist.org/k/sHbOUOi_7BG68qUDIbeTbQ/bQTi5?s=edit              baltimor
e | create posting                                                                 1
12  https://post.craigslist.org/k/sHbOUOi_7BG68qUDIbeTbQ/bQTi5?s=geoverify         baltimor
e | add map                                                                        1
13  https://post.craigslist.org/k/sHbOUOi_7BG68qUDIbeTbQ/bQTi5?s=editimage         baltimor
e | create posting                                                                 1
14  https://post.craigslist.org/k/sHbOUOi_7BG68qUDIbeTbQ/bQTi5?s=preview           baltimor
e | create posting                                                                 1
15  https://post.craigslist.org/manage/7473121658                                  baltimor
e | manage posting                                                                 1
16  https://baltimore.craigslist.org/hab/d/baltimore-cheaper-than-rx-supplements/747  cheaper
than Rx supplements - health and beauty - by owner -  1
```

Figure 4.6 — First 20 rows of urls (id, url, title, visit_count), confirming the Craigslist posting and Gmail URLs are present

```
16  https://baltimore.craigslist.org/hab/d/baltimore-cheaper-than-rx-supplements/747  cheaper
than Rx supplements - health and beauty - by owner -  1
17  https://www.google.com/search?q=gmail&rlz=1C1ONGR_enUS989US989&oq=gmail&aqs=chro  gmail -
Google Search                                                                      2
18  https://www.google.com/gmail/                                                  Gmail: F
ree, Private & Secure Email | Google Workspace                                     1
19  https://mail.google.com/mail/                                                  Inbox (1
2) - unsub.fscs@gmail.com - Gmail                                                  2
20  https://mail.google.com/mail/u/0/                                              Inbox (1
2) - unsub.fscs@gmail.com - Gmail                                                  2
```

Figure 4.7 — Continuation of the urls listing, rows 16–20: Gmail and blockchain-related entries

```
┌──(kali㉿kali)-[~/SBT-DF204-Case2/working]
└─$ sqlite3 History_working -header -column "SELECT * FROM keyword_search_terms;"
keyword_id  url_id  term                normalized_term
----------  ------  ------------------  ------------------
2           1       craigslist          craigslist
2           17      gmail               gmail
2           40      blockchain explorer  blockchain explorer
2           45      kraken              kraken
```

Figure 4.8 — keyword_search_terms table: craigslist, gmail, blockchain explorer, kraken — the four omnibox searches typed by the user

```
┌──(kali㉿kali)-[~/SBT-DF204-Case2/working]
└─$ sqlite3 History_working -header -column "SELECT id, current_path, target_path, start_time,
 received_bytes, total_bytes, mime_type FROM downloads;"
id  current_path                                target_path
        start_time        received_bytes  total_bytes  mime_type
--  ------------------------------------------  ------------------------------------------
------  ----------------  --------------  -----------  ----------
1   C:\Users\FSCS_User\Desktop\proof_of_payment.png  C:\Users\FSCS_User\Desktop\proof_of_payme
nt.png  13294853818494344  33844           33844        image/png
```

Figure 4.9 — downloads table: the single record for proof_of_payment.png (33,844 bytes, image/png)

Timestamp conversion was verified with the date-range and timezone checks shown below. The session spans 2022-04-19 13:54:18 UTC to 2022-04-19 15:02:56 UTC, a span of exactly 4,118 seconds across 74 visits and 57 distinct URLs.

```
┌──(kali㉿kali)-[~/SBT-DF204-Case2/working]
└─$ sqlite3 History_working -header -column "
SELECT
  datetime(MIN(v.visit_time)/1000000 + strftime('%s','1601-01-01'), 'unixepoch') AS first_visit_utc,
  datetime(MAX(v.visit_time)/1000000 + strftime('%s','1601-01-01'), 'unixepoch') AS last_visit_utc,
  COUNT(*) AS total_visits,
  (MAX(v.visit_time) - MIN(v.visit_time))/1000000 AS span_seconds
FROM visits v;
"
first_visit_utc      last_visit_utc       total_visits  span_seconds
-------------------  -------------------  ------------  ------------
2022-04-19 13:54:18  2022-04-19 15:02:56  74            4118
```

Figure 4.10 — MIN/MAX visit_time converted to UTC: first_visit_utc = 2022-04-19 13:54:18, last_visit_utc = 2022-04-19 15:02:56, total_visits = 74, span_seconds = 4118

```
┌──(kali㉿kali)-[~/SBT-DF204-Case2/working]
└─$ timedatectl | grep -i "Time zone"
               Time zone: America/New_York (EDT, -0400)
```

Figure 4.11 — Analysis-workstation timezone check: timedatectl reports America/New_York (EDT, -0400)

```
┌──(kali㉿kali)-[~/SBT-DF204-Case2/working]
└─$ sqlite3 History_working -header -column "
SELECT v.id AS visit_id,
       datetime(v.visit_time/1000000 + strftime('%s','1601-01-01'), 'unixepoch') AS visit_utc,
       datetime(v.visit_time/1000000 + strftime('%s','1601-01-01'), 'unixepoch', 'localtime') AS visit_local
FROM visits AS v
ORDER BY v.visit_time
LIMIT 5;
"
visit_id  visit_utc            visit_local
--------  -------------------  -------------------
1         2022-04-19 13:54:18  2022-04-19 09:54:18
2         2022-04-19 13:54:19  2022-04-19 09:54:19
3         2022-04-19 13:54:19  2022-04-19 09:54:19
4         2022-04-19 13:54:19  2022-04-19 09:54:19
5         2022-04-19 13:54:19  2022-04-19 09:54:19
```

Figure 4.12 — Side-by-side UTC and local-time conversion of the first five visits, confirming a consistent −4-hour offset (13:54:18 UTC = 09:54:18 local)

All timestamps in this report are given in UTC, as recommended in the assignment, because UTC is unambiguous and independent of the analysis workstation's configured timezone. Where a local-time equivalent is useful for context it is noted separately and is always the UTC value minus 4 hours (America/New_York, EDT, UTC−4:00, confirmed via timedatectl on the analysis VM) — this is the analysis workstation's local timezone used for display purposes only, and is not a determination of the suspect device's own configured timezone, which the History database does not record.

Evidence reference: E02a–i, E03a–d.

5. Findings

Each investigation question from the assignment brief is addressed below with record IDs, SQL output, and the corresponding screenshot evidence.

5.1 Investigation Question 4 — How did the browser reach the material URLs?

Transition decoding followed the method given in the brief: base type = transition AND 0xFF, with qualifier bits read from the remaining high-order bits (CHAIN_START = 0x10000000, CLIENT_REDIRECT = 0x20000000, SERVER_REDIRECT = 0x40000000, and a fourth high bit used by the vendor dissector for FORWARD_BACK/API-driven navigation). Grouping all 74 visits by (transition & 0xFF) produced the distribution below.

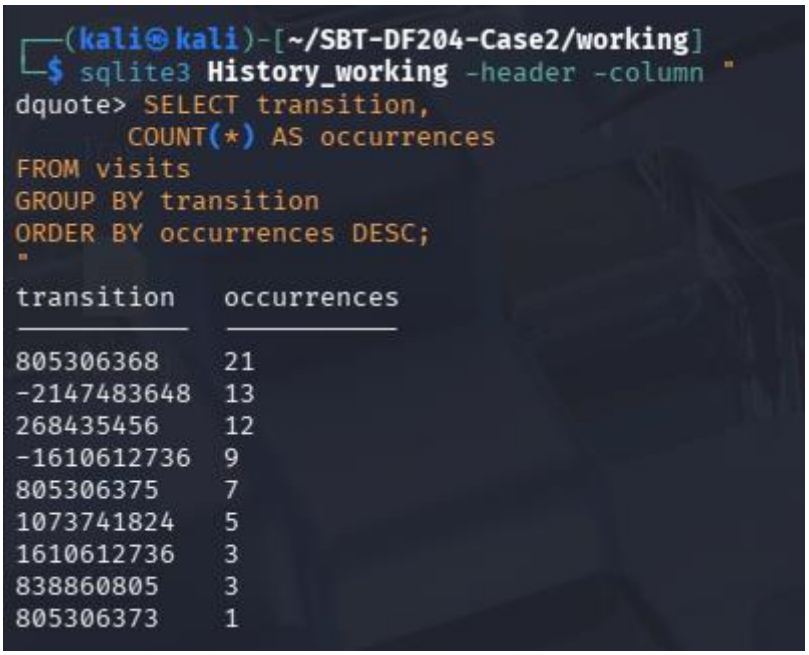


Figure 5.1 — Base-transition distribution: LINK (click) 63, FORM_SUBMIT (filled form) 7, OMNIBOX (typed search) 4

Base transition	Occurrences	Interpretation
0 — LINK	63 (85.1%)	Followed a hyperlink already present on a page
7 — FORM_SUBMIT	7 (9.5%)	Submitted an HTML form (all 7 fall within the Craigslist posting workflow)
5 — OMNIBOX / GENERATED	4 (5.4%)	Typed a search term into the Chrome address bar (craigslist, gmail, blockchain explorer, kraken)

Base transition	Occurrences	Interpretation
1 — TYPED	0 (0%)	No visits in this session began from a directly typed URL



Figure 5.2 — Python decode of the 9 distinct raw transition values into base type plus qualifier flags (CHAIN_START, CLIENT_REDIRECT, SERVER_REDIRECT, FORWARD_BACK)

Nine distinct raw transition values were observed across the 74 visits. The most frequent single value, 0x30000007 (FORM_SUBMIT + CHAIN_START + CLIENT_REDIRECT), accounts for the seven-step Craigslist posting workflow. No TYPED (base = 1) transitions exist anywhere in the session — the user never typed a full URL directly; every navigation originated from a link, a filled form, or a typed search term that was then clicked from the results.

```
┌──(kali㉿kali)-[~/SBT-DF204-Case2/working]
└─$ sqlite3 History_working -header -column "
SELECT v.id AS visit_id,
       u.id AS url_id,
       u.url,
       u.title,
       datetime(v.visit_time/1000000 + strftime('%s','1601-01-01'), 'unixepoch') AS visit_utc,

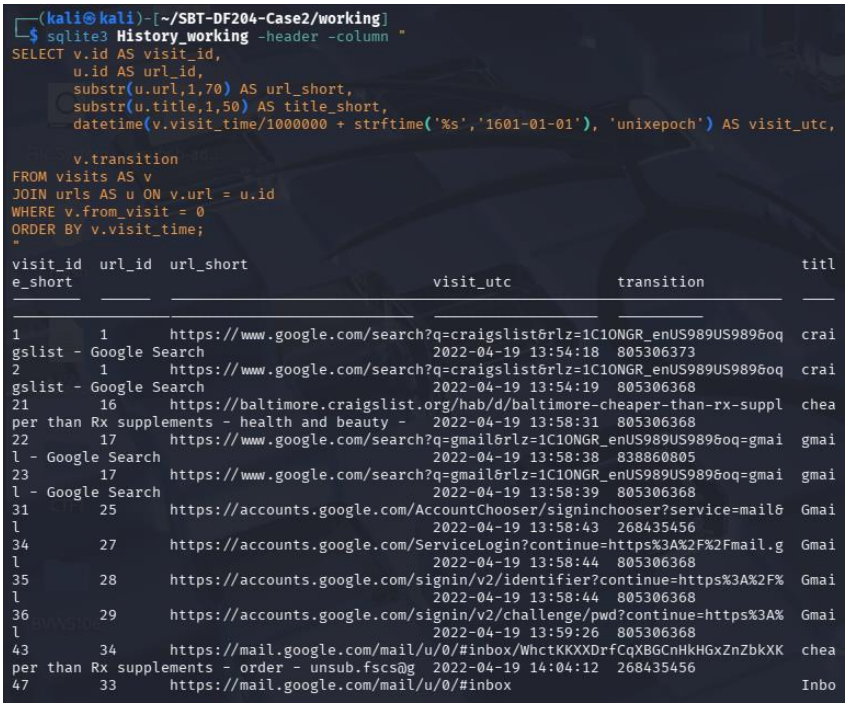
       v.from_visit,
       v.transition
FROM visits AS v
JOIN urls AS u ON v.url = u.id
WHERE (v.transition & 0×FF) = 7
ORDER BY v.visit_time;
"
visit_id  url_id  url                                                                titl
e                                  visit_utc            from_visit  transition
────────  ──────  ─────────────────────────────────────  ───────────────────  ──────────  ──────────
8         5       https://accounts.craigslist.org/login/home                         crai
gslist account             2022-04-19 13:54:39  7           805306375
14        10      https://post.craigslist.org/k/sHbOUOi_7BG68qUDIbeTbQ/bQTi5?s=cat   balt
imore | choose category    2022-04-19 13:55:10  13          805306375
15        11      https://post.craigslist.org/k/sHbOUOi_7BG68qUDIbeTbQ/bQTi5?s=edit  balt
imore | create posting     2022-04-19 13:55:15  14          805306375
16        12      https://post.craigslist.org/k/sHbOUOi_7BG68qUDIbeTbQ/bQTi5?s=geoverify  balt
imore | add map            2022-04-19 13:58:13  15          805306375
17        13      https://post.craigslist.org/k/sHbOUOi_7BG68qUDIbeTbQ/bQTi5?s=editimage  balt
imore | create posting     2022-04-19 13:58:17  16          805306375
18        14      https://post.craigslist.org/k/sHbOUOi_7BG68qUDIbeTbQ/bQTi5?s=preview  balt
imore | create posting     2022-04-19 13:58:19  17          805306375
19        8       https://post.craigslist.org/k/sHbOUOi_7BG68qUDIbeTbQ/bQTi5          balt
imore | posting confirmation  2022-04-19 13:58:22  18          805306375
```

Figure 5.3 — Query isolating FORM_SUBMIT visits ((transition & 0xFF) = 7): the 7-step Craigslist posting sequence, visit IDs 8 and 14–19



Figure 5.4 — Query isolating OMNIBOX visits ((transition & 0xFF) = 5): the four typed searches — craigslist, gmail, blockchain explorer, kraken — each followed by a results-page click

from_visit chaining was used to reconstruct navigation sequences: a from_visit of 0 marks the start of a new chain (typically a typed search, a bookmark, or a link opened from outside the captured session, e.g. from an external email). 22 of the 74 visits have from_visit = 0, marking the start points of the distinct activity threads examined in Sections 5.2–5.3 below (the Craigslist search, the Gmail order thread, the txid-referencing email, the Imgur redirect, the blockchain.com search, and the Kraken searches each begin their own chain).



```
┌──(kali㉿kali)-[~/SBT-DF204-Case2/working]
└─$ sqlite3 History_working -header -column "
SELECT v.id AS visit_id,
       u.id AS url_id,
       substr(u.url,1,70) AS url_short,
       substr(u.title,1,50) AS title_short,
       datetime(v.visit_time/1000000 + strftime('%s','1601-01-01'), 'unixepoch') AS visit_utc,

       v.transition
FROM visits AS v
JOIN urls AS u ON v.url = u.id
WHERE v.from_visit = 0
ORDER BY v.visit_time;
"
visit_id  url_id  url_short                                                
```

a Google search for "craigslist", account login, and a complete seven-step posting workflow.

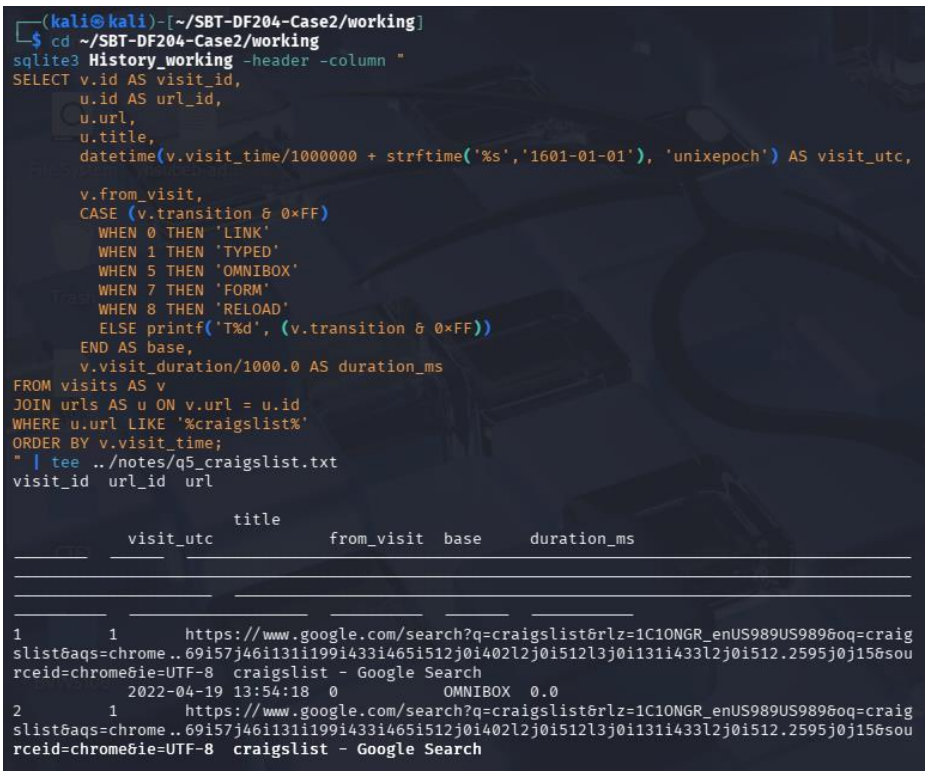


Figure 5.7 — Query isolating all 21 craigslist.org-domain visits with decoded base transition

Visit ID	UTC time	URL / stage	Base
1–2	13:54:18–19	Google search for "craigslist" (OMNIBOX, then the results-page click)	OMNIBOX / LINK
3–5	13:54:19	craigslist.org → geo.craigslist.org → baltimore.craigslist.org (geographic redirect chain)	LINK
6–8	13:54:22–39	accounts.craigslist.org login/home → login form → FORM_SUBMIT back to /login/home	LINK / FORM
9–11	13:55:03–05	Return to craigslist.org → baltimore.craigslist.org → post.craigslist.org/c/bal (choose posting location)	LINK
12–19	13:55:05–13:58:22	post.craigslist.org/k/… posting token: choose type → choose category (FORM) → create posting (FORM) → add map (FORM) → edit image (FORM) → preview (FORM) → posting confirmation (FORM)	LINK / FORM
20	13:58:27	post.craigslist.org/manage/7473121658 — author-only "manage posting" page	LINK
21	13:58:31	baltimore.craigslist.org/…/baltimore-cheaper-than-rx-supplements/7473121658.html — the live published ad	LINK



Figure 5.8 — Visit 20 in detail: url_id 15, https://post.craigslist.org/manage/7473121658, title "baltimore | manage posting", 2022-04-19 13:58:27 UTC



Figure 5.9 — Visit 21 in detail: url_id 16, the published ad at /hab/d/baltimore-cheaper-than-rx-supplements/7473121658.html, 2022-04-19 13:58:31 UTC



Figure 5.10 — Direct match query: both the /manage/7473121658 URL and the published ad URL share posting ID 7473121658

Correlation: the numeric posting ID 7473121658 is identical between the author-only management URL (visit 20, url_id 15) and the published advertisement URL (visit 21, url_id 16, titled "cheaper than Rx supplements - health and beauty - by owner - ..."). The /manage/ path on post.craigslist.org is only reachable by the account that created the posting, so this match ties the browser profile that completed the login and seven-

step FORM_SUBMIT workflow (visits 6–19) directly to authorship of the published ad.



Figure 5.11 — Full-field record for visit 21: the published ad URL and title, confirming visit_count = 1 and the exact UTC timestamp 2022-04-19 13:58:31

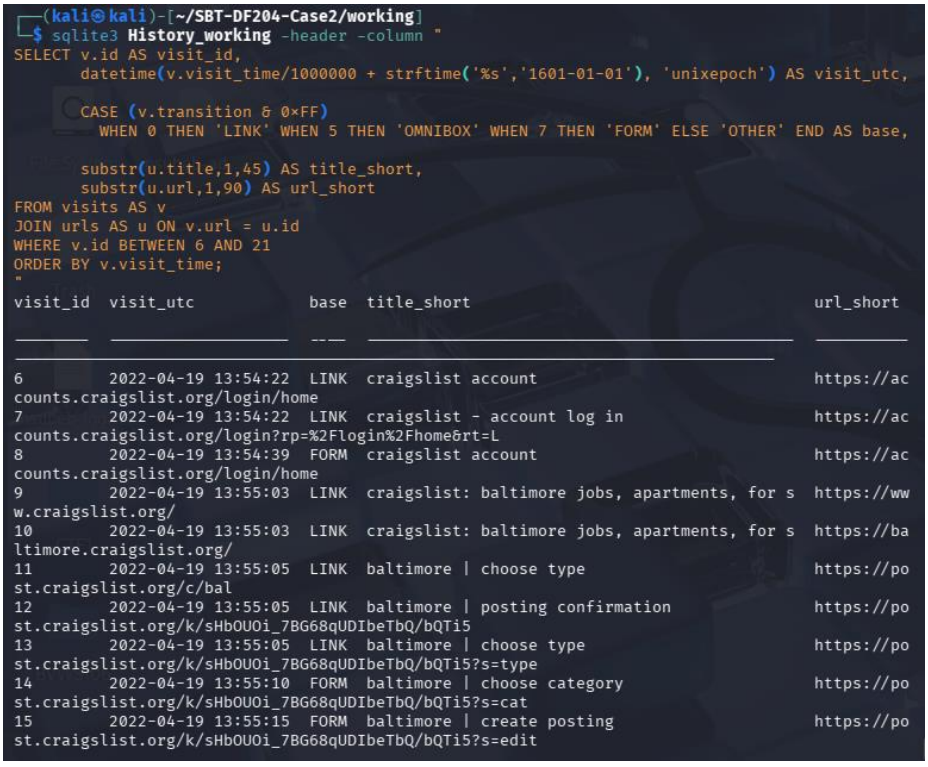
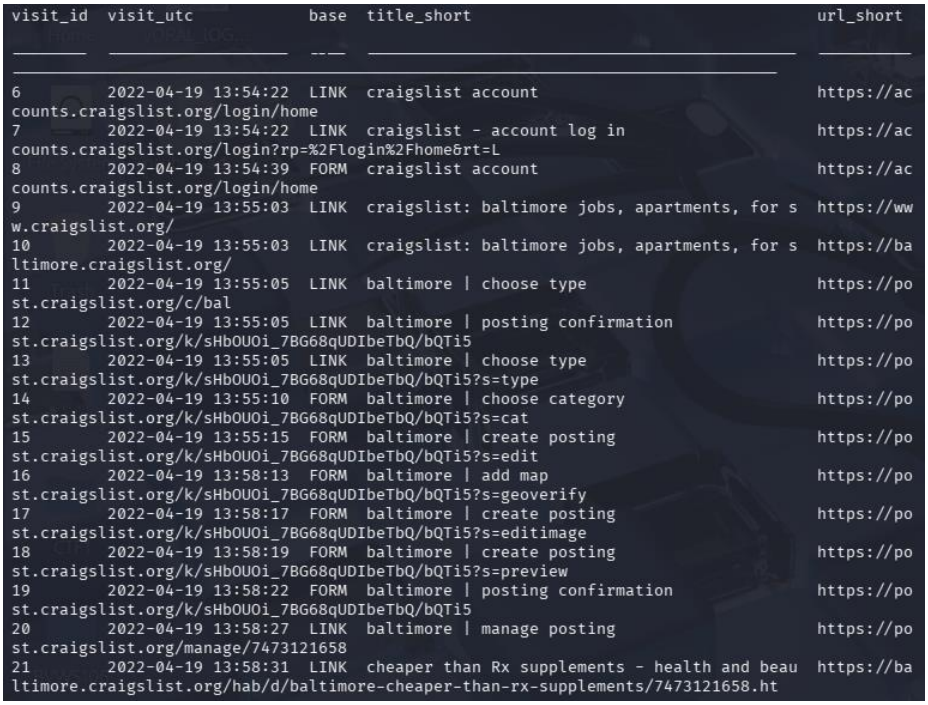


Figure 5.12 — Visits 6–15 of the Craigslist thread with decoded base transitions: account login through "create posting"



visit_id	visit_utc	base	title_short	url_short
6	2022-04-19 13:54:22	LINK	craigslist account	https://accounts.craigslist.org/login/home
7	2022-04-19 13:54:22	LINK	craigslist - account log in	https://accounts.craigslist.org/login?rp=%2Flogin%2Fhome&rt=L
8	2022-04-19 13:54:39	FORM	craigslist account	https://accounts.craigslist.org/login/home
9	2022-04-19 13:55:03	LINK	craigslist: baltimore jobs, apartments, for s	https://www.craigslist.org/
10	2022-04-19 13:55:03	LINK	craigslist: baltimore jobs, apartments, for s	https://baltimore.craigslist.org/
11	2022-04-19 13:55:05	LINK	baltimore | choose type	https://post.craigslist.org/c/bal
12	2022-04-19 13:55:05	LINK	baltimore | posting confirmation	https://post.craigslist.org/k/sHbOUOi_7BG68qUDIbeTbQ/bQTi5
13	2022-04-19 13:55:05	LINK	baltimore | choose type	https://post.craigslist.org/k/sHbOUOi_7BG68qUDIbeTbQ/bQTi5?s=type
14	2022-04-19 13:55:10	FORM	baltimore | choose category	https://post.craigslist.org/k/sHbOUOi_7BG68qUDIbeTbQ/bQTi5?s=cat
15	2022-04-19 13:55:15	FORM	baltimore | create posting	https://post.craigslist.org/k/sHbOUOi_7BG68qUDIbeTbQ/bQTi5?s=edit
16	2022-04-19 13:58:13	FORM	baltimore | add map	https://post.craigslist.org/k/sHbOUOi_7BG68qUDIbeTbQ/bQTi5?s=geoverify
17	2022-04-19 13:58:17	FORM	baltimore | create posting	https://post.craigslist.org/k/sHbOUOi_7BG68qUDIbeTbQ/bQTi5?s=editimage
18	2022-04-19 13:58:19	FORM	baltimore | create posting	https://post.craigslist.org/k/sHbOUOi_7BG68qUDIbeTbQ/bQTi5?s=preview
19	2022-04-19 13:58:22	FORM	baltimore | posting confirmation	https://post.craigslist.org/k/sHbOUOi_7BG68qUDIbeTbQ/bQTi5
20	2022-04-19 13:58:27	LINK	baltimore | manage posting	https://post.craigslist.org/manage/7473121658
21	2022-04-19 13:58:31	LINK	cheaper than Rx supplements - health and beau	https://baltimore.craigslist.org/hab/d/baltimore-cheaper-than-rx-supplements/7473121658.ht

Figure 5.13 — Continuation, visits 16–21: "add map" through the published ad, FORM qualifiers 0x30000007 throughout the posting steps

Inference: the browser profile searched for Craigslist, logged into an account, and completed a seven-stage posting form culminating in a live advertisement titled "cheaper than Rx supplements" under a health-and-beauty category. Limitation: the database records only URLs, titles and transition metadata — it does not capture the text entered into the posting form, so the ad's exact wording, price and images are known only from the title string and are not independently verifiable from this database alone.

Evidence reference: E05a–f.

5.3 Investigation Question 6 — What evidence relates to communications and payment activity?

5.3.1 Gmail account and the "order" thread

25 of the 74 visits are to mail.google.com. Six page titles contain the literal account identifier "Inbox (12) - unsub.fscs@gmail.com - Gmail", directly naming the Gmail account active in this browser profile throughout the session.

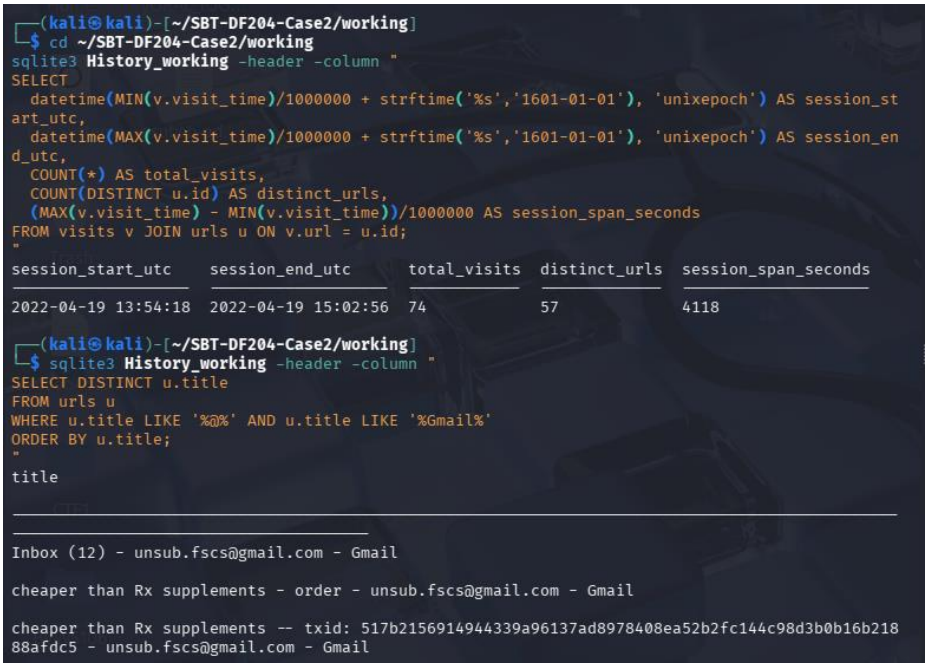


Figure 5.14 — DISTINCT title query for '%@%' AND '%Gmail%': the account unsub.fscs@gmail.com appears in the inbox title and in both the order-thread and txid-thread subject lines

Visit ID	UTC time	Subject / page	from_visit	Transition
43	14:04:12	"cheaper than Rx supplements - order - unsub.fscs@gmail.com - Gmail" (thread opened)	0	LINK (chain start)
44	14:04:35	Same thread, compose/reply opened	43	CLIENT_REDIRECT
45	14:04:46	Same thread, second compose/reply opened	44	CLIENT_REDIRECT
46	14:04:52	Thread view (reply closed)	45	FORM / redirect (0x60000000)
47	14:05:59	Return to inbox (last visit before the 50m48s gap)	0	LINK

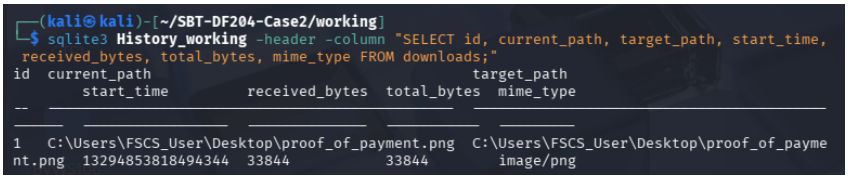


Figure 5.15 — (repeated for context) the retained download, proof_of_payment.png, 33,844 bytes, sourced from tab_url https://imgur.com/dTgrkP7

Inference: the "order" email thread (visit IDs 43–46) was opened and replied to twice (two separate compose events, visits 44 and 45) within the same browsing session that produced the Craigslist posting, roughly 5–6 minutes after the ad went live (13:58:31 → 14:04:12 UTC).

5.3.2 The transaction-ID email and the 50-minute activity gap

After visit 47 (14:05:59 UTC), the next recorded visit (48) does not occur until 14:56:47 UTC — a gap of 50 minutes 48 seconds, the single largest idle period in the entire 74-visit session.



Figure 5.16 — Visits 46–49 isolated: the gap sits between visit 47 (14:05:59) and visit 48 (14:56:47)

Visit 48, the first activity after the gap, opens a Gmail thread whose page title embeds a Bitcoin transaction ID verbatim: “cheaper than Rx supplements -- txid: 517b2156914944339a96137ad8978408ea52b2fc144c98d3b0b16b21888afdc5 - unsub.fscs@gmail.com - Gmail”.

Visit ID	UTC time	Subject	from_visit
48	14:56:47	"cheaper than Rx supplements -- txid: 517b2156…afdc5" (thread opened, chain start)	0
72	15:02:32	Same thread, reply/compose opened	48
73	15:02:44	Same thread, second reply/compose opened	72
74	15:02:56	Thread view (final visit of the session)	73

5.3.3 Imgur proof-of-payment image

Five seconds after the txid email was opened (14:56:52 UTC), the browser followed a Gmail-generated redirect (google.com/url?q=…&source=gmail&…) to an Imgur page, https://imgur.com/dTgrkP7, titled “Imgur: The magic of the Internet”, where it remained for 18.9 seconds (visit_duration ≈ 18,926 ms) before the next navigation.

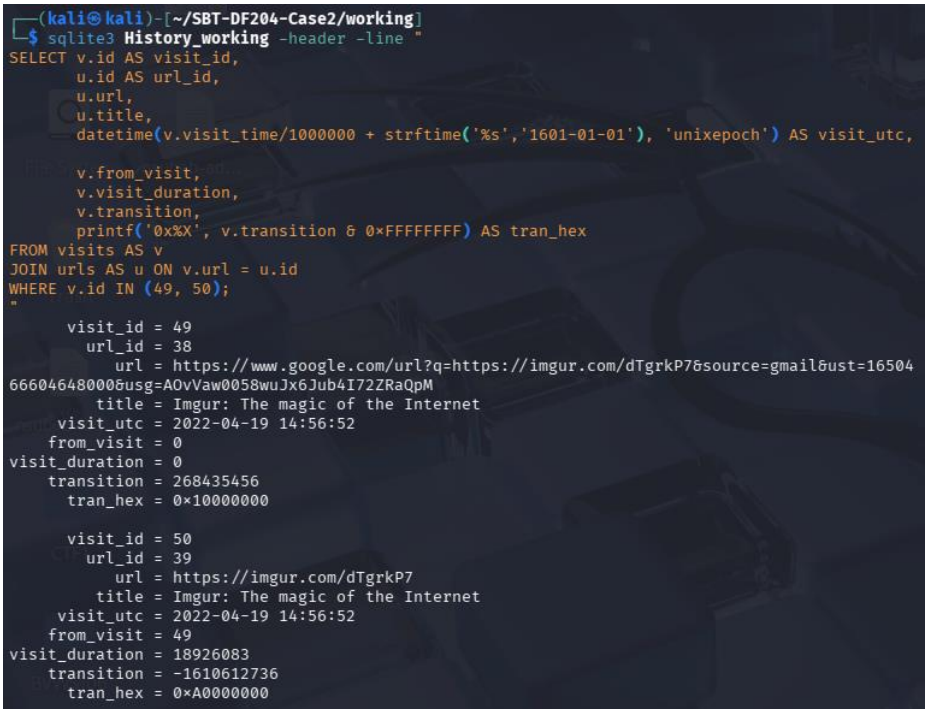


Figure 5.17 — Visits 49–50 in full detail: the gmail-sourced redirect to imgur.com/dTgrkP7, with an 18.9-second dwell time on the Imgur page

This Imgur URL matches the tab_url field of the single downloads record (download id = 1): a 33,844-byte PNG, saved locally as C:\Users\FSCS_User\Desktop\proof_of_payment.png, downloaded between 2022-04-19 14:56:58 and 14:57:10 UTC — i.e. within seconds of the browser visiting the Imgur page it was sourced from. The downloaded and total byte counts are equal (33,844 = 33,844), confirming a complete, uninterrupted download rather than a partial or failed transfer.

5.3.4 Blockchain.com transaction verification

The same transaction ID from the txid email (517b2156…afdc5) was then looked up directly on blockchain.com, via both its site search and the direct /btc/tx/ permalink.

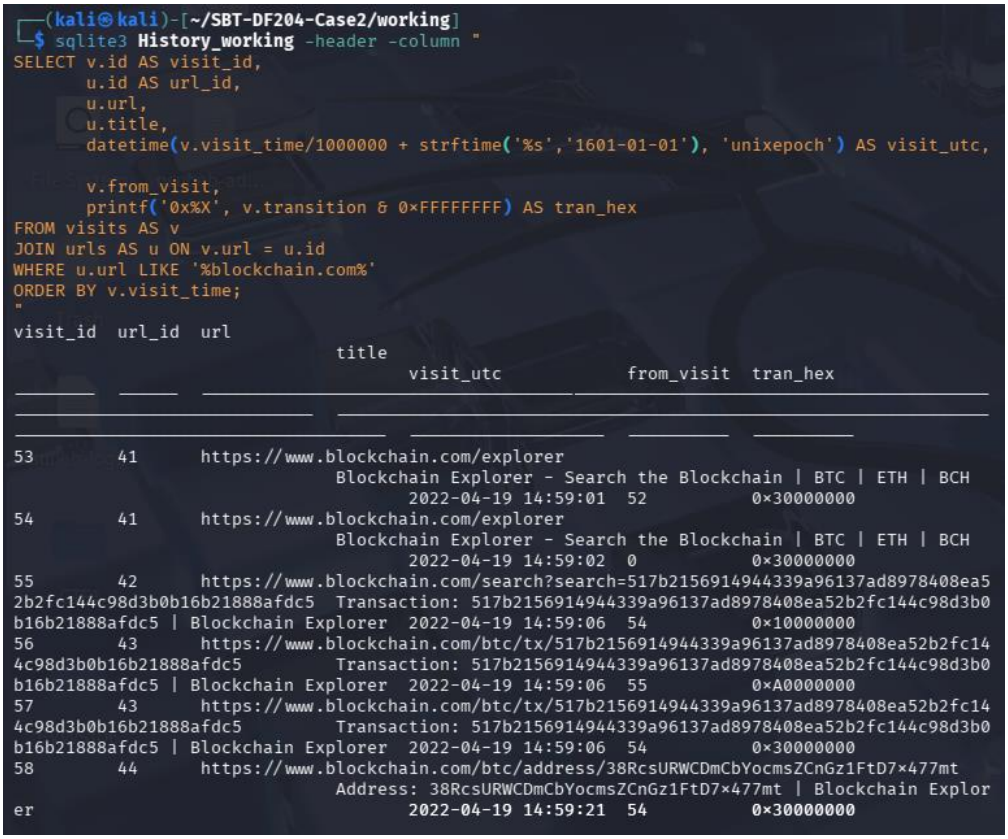


Figure 5.18 — Visits 53–58: blockchain.com/explorer search, the txid search-results page, the /btc/tx/ permalink (visited twice), and a lookup of wallet address 38RcsURWCDmCbYocmsZCnGz1FtD7x477mt

Visit ID	UTC time	URL	Artefact
55	14:59:06	blockchain.com/search?search=517b2156…afdc5	Site-search for the exact transaction ID
56–57	14:59:06	blockchain.com/btc/tx/517b2156…afdc5	Transaction permalink opened (visited twice in the same second — consistent with a results-page redirect)
58	14:59:21	blockchain.com/btc/address/38RcsURWCDmCbYocmsZCnGz1FtD7x477mt	Lookup of the wallet address associated with the transaction

5.3.5 Kraken exchange activity and a second, independent txid confirmation

At 14:59:38 UTC the user typed “kraken” into the omnibox, signed in (14:59:47), and approved a new-device prompt (15:00:21) — consistent with accessing an exchange account from a browser profile not previously trusted by Kraken.

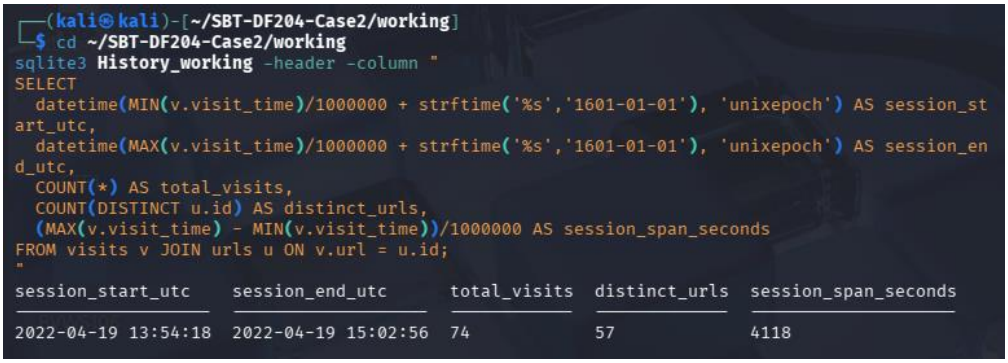


Figure 5.19 — Kraken session visits 61–71: sign-in, device approval, trade/instant/funding pages, and a history/ledger page

Visit ID	UTC time	URL	Note
61–62	14:59:47	kraken.com/en-us/sign-in	Sign-in page
63	15:00:21	kraken.com/en-us/device-approval	New-device approval step
64–65	15:01:11–12	kraken.com/u/trade, kraken.com/u/instant	Trading pages
66	15:01:21	kraken.com/u/funding — title "41,543.80 USD - Kraken - Funding - Overview"	First balance figure visible in the page title
67–70	15:01:31–33	kraken.com/redirect…mempool.space/tx/517b2156…afdc5 → mempool.space/tx/517b2156…afdc5	Kraken-hosted redirect to mempool.space, landing on the SAME transaction ID, a third source confirming it
71	15:02:05	kraken.com/u/history/ledger — title "41,563.70 USD - Kraken - History - Ledger"	Second balance figure (41,563.70, 19.90 higher than the funding-page figure 6m44s earlier)

Correlation: the transaction ID 517b2156914944339a96137ad8978408ea52b2fc144c98d3b0b16b21888afdc5 appears, verbatim, in nine records across four distinct platforms and contexts within this single database: (1) the Gmail email-thread title (visit 48, and again 72–74), (2) the blockchain.com search and permalink (visits 55–57), (3) the Kraken-to-mempool.space redirect chain (visits 67–70), and (4) the mempool.space transaction page reached via that redirect. Cross-referencing the same unique identifier through four independently operated services within one continuous session is a deliberate verification pattern, not a coincidental one.

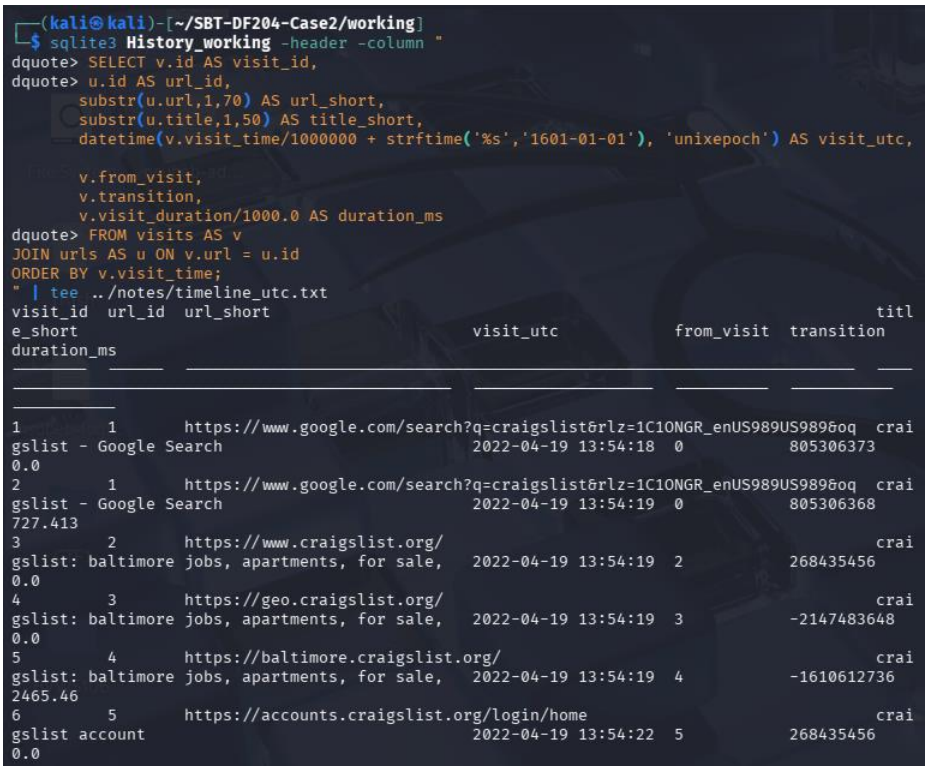
Limitation: the two Kraken balance figures (41,543.80 USD at 15:01:21 and 41,563.70 USD at 15:02:05, a difference of 19.90 USD) are read only from page titles, not from a captured account statement, and the database does not record whether this difference reflects the transaction under

investigation, an unrelated balance movement, or a display/exchange-rate artefact. This should be treated as a suggestive observation, not a proven payment amount.

Evidence reference: E06a–j.

6. Timeline (Investigation Question 7)

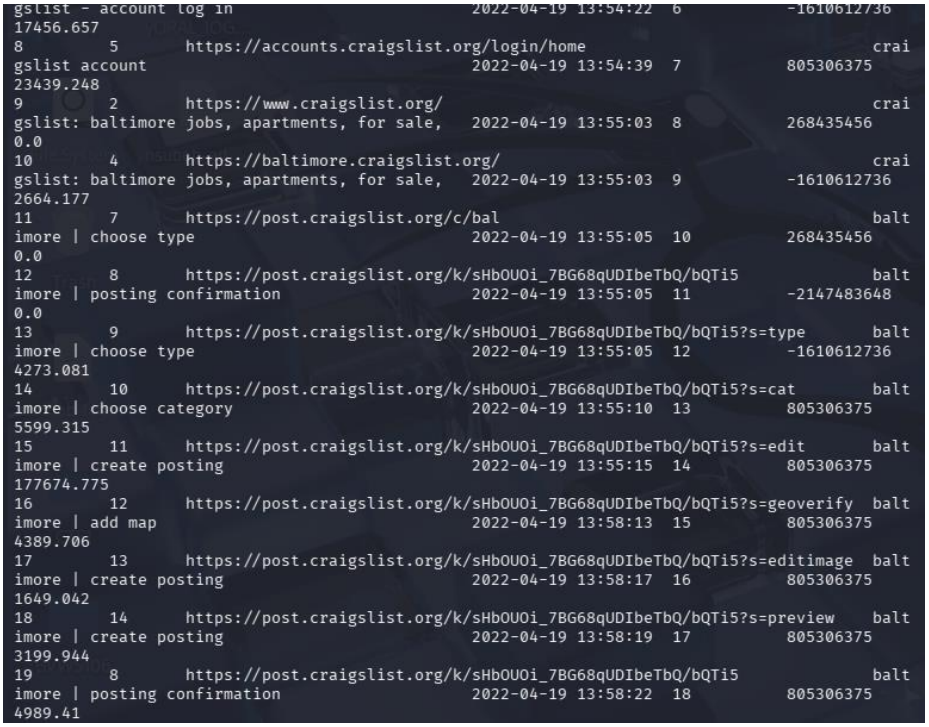
Investigation question: What is the reconstructed timeline? All timestamps below are UTC, converted from the Chrome visit_time field using datetime(visit_time/1000000 + strftime('%s','1601-01-01'), 'unixepoch'). The session covers 2022-04-19 13:54:18 – 15:02:56 UTC, 74 visits, 57 distinct URLs, 4,118 seconds total span. The table separates the observed artefact from this report's interpretation of the activity stage, as required by the assignment brief.



```
┌──(kali㉿kali)-[~/SBT-DF204-Case2/working]
└─$ sqlite3 History_working -header -column "
dquote> SELECT v.id AS visit_id,
dquote> u.id AS url_id,
       substr(u.url,1,70) AS url_short,
       substr(u.title,1,50) AS title_short,
       datetime(v.visit_time/1000000 + strftime('%s','1601-01-01'), 'unixepoch') AS visit_utc,

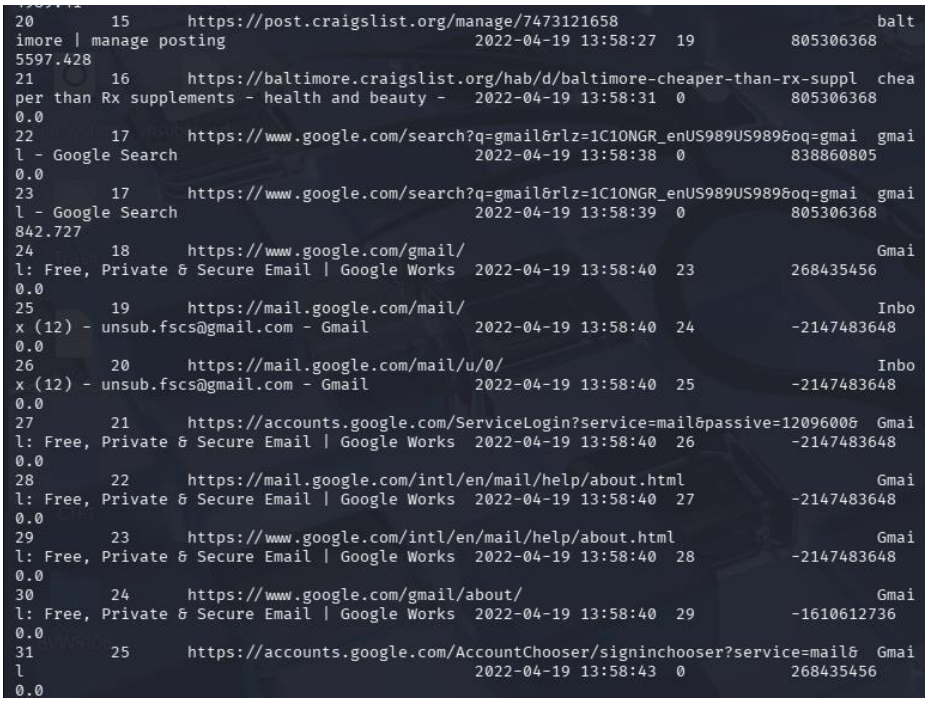
       v.from_visit,
       v.transition,
       v.visit_duration/1000.0 AS duration_ms
dquote> FROM visits AS v
JOIN urls AS u ON v.url = u.id
ORDER BY v.visit_time;
" | tee ../notes/timeline_utc.txt
visit_id  url_id  url_short                                                        titl
e_short                                  visit_utc            from_visit  transition
duration_ms
────────  ──────  ──────────────────────────────────────────────────────────────  ───
──────────────────────────────────────  ───────────────────  ──────────  ──────────
──────────
1         1       https://www.google.com/search?q=craigslist&rlz=1C1ONGR_enUS989US989&oq  crai
gslist - Google Search                   2022-04-19 13:54:18  0           805306373
0.0
2         1       https://www.google.com/search?q=craigslist&rlz=1C1ONGR_enUS989US989&oq  crai
gslist - Google Search                   2022-04-19 13:54:19  0           805306368
727.413
3         2       https://www.craigslist.org/                                     crai
gslist: baltimore jobs, apartments, for sale,    2022-04-19 13:54:19  2           268435456
0.0
4         3       https://geo.craigslist.org/                                     crai
gslist: baltimore jobs, apartments, for sale,    2022-04-19 13:54:19  3           -2147483648
0.0
5         4       https://baltimore.craigslist.org/                               crai
gslist: baltimore jobs, apartments, for sale,    2022-04-19 13:54:19  4           -1610612736
2465.46
6         5       https://accounts.craigslist.org/login/home                      crai
gslist account                           2022-04-19 13:54:22  5           268435456
0.0
```

Figure 6.1 — Full UTC-converted timeline, visits 1–6: the Craigslist search and geo-redirect chain



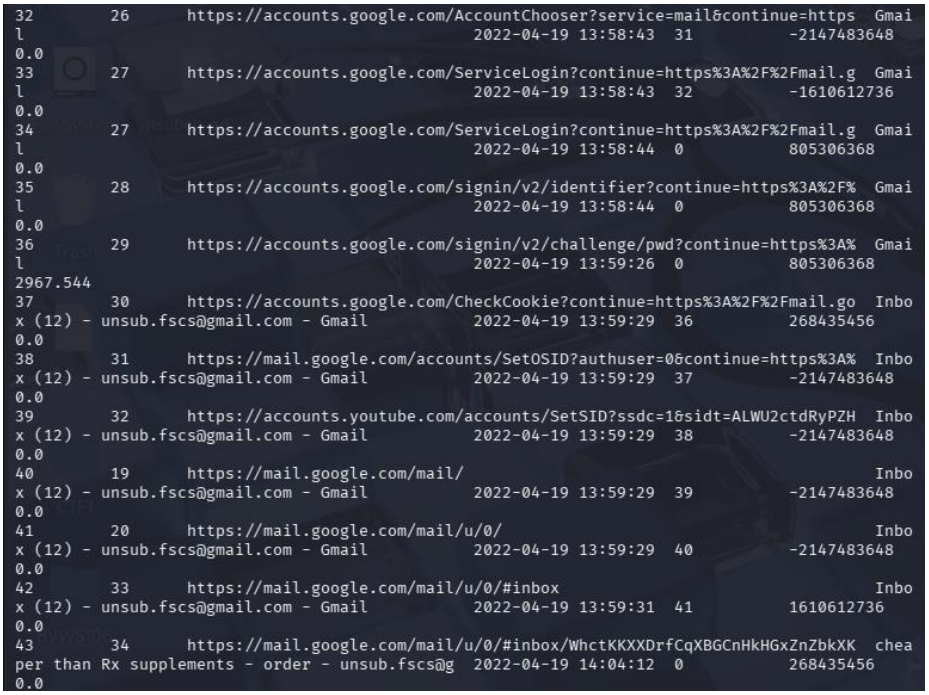
gslist - account log in			2022-04-19 13:54:22	6		-1610612736	17456.657
8	5	https://accounts.craigslist.org/login/home				crai	
gslist account			2022-04-19 13:54:39	7		805306375	23439.248
9	2	https://www.craigslist.org/				crai	
gslist: baltimore jobs, apartments, for sale,			2022-04-19 13:55:03	8		268435456	0.0
10	4	https://baltimore.craigslist.org/				crai	
gslist: baltimore jobs, apartments, for sale,			2022-04-19 13:55:03	9		-1610612736	2664.177
11	7	https://post.craigslist.org/c/bal				balt	
imore | choose type			2022-04-19 13:55:05	10		268435456	0.0
12	8	https://post.craigslist.org/k/sHbOUOi_7BG68qUDIbeTbQ/bQTi5				balt	
imore | posting confirmation			2022-04-19 13:55:05	11		-2147483648	0.0
13	9	https://post.craigslist.org/k/sHbOUOi_7BG68qUDIbeTbQ/bQTi5?s=type				balt	
imore | choose type			2022-04-19 13:55:05	12		-1610612736	4273.081
14	10	https://post.craigslist.org/k/sHbOUOi_7BG68qUDIbeTbQ/bQTi5?s=cat				balt	
imore | choose category			2022-04-19 13:55:10	13		805306375	5599.315
15	11	https://post.craigslist.org/k/sHbOUOi_7BG68qUDIbeTbQ/bQTi5?s=edit				balt	
imore | create posting			2022-04-19 13:55:15	14		805306375	177674.775
16	12	https://post.craigslist.org/k/sHbOUOi_7BG68qUDIbeTbQ/bQTi5?s=geoverify				balt	
imore | add map			2022-04-19 13:58:13	15		805306375	4389.706
17	13	https://post.craigslist.org/k/sHbOUOi_7BG68qUDIbeTbQ/bQTi5?s=editimage				balt	
imore | create posting			2022-04-19 13:58:17	16		805306375	1649.042
18	14	https://post.craigslist.org/k/sHbOUOi_7BG68qUDIbeTbQ/bQTi5?s=preview				balt	
imore | create posting			2022-04-19 13:58:19	17		805306375	3199.944
19	8	https://post.craigslist.org/k/sHbOUOi_7BG68qUDIbeTbQ/bQTi5				balt	
imore | posting confirmation			2022-04-19 13:58:22	18		805306375	4989.41

Figure 6.2 — Continuation: the login and early posting-workflow steps



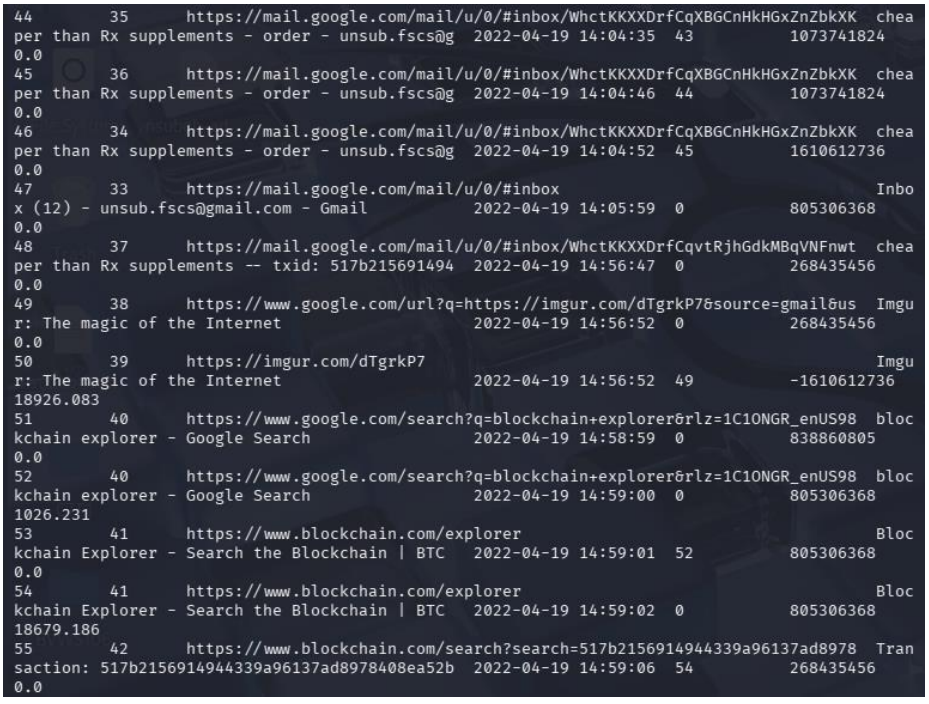
20	15	https://post.craigslist.org/manage/7473121658				balt	
imore | manage posting			2022-04-19 13:58:27	19		805306368	5597.428
21	16	https://baltimore.craigslist.org/hab/d/baltimore-cheaper-than-rx-suppl				chea	
per than Rx supplements - health and beauty -			2022-04-19 13:58:31	0		805306368	0.0
22	17	https://www.google.com/search?q=gmail&rlz=1C1ONGR_enUS989US989&oq=gmai				gmai	
l - Google Search			2022-04-19 13:58:38	0		838860805	0.0
23	17	https://www.google.com/search?q=gmail&rlz=1C1ONGR_enUS989US989&oq=gmai				gmai	
l - Google Search			2022-04-19 13:58:39	0		805306368	842.727
24	18	https://www.google.com/gmail/				Gmai	
l: Free, Private & Secure Email | Google Works			2022-04-19 13:58:40	23		268435456	0.0
25	19	https://mail.google.com/mail/				Inbo	
x (12) - unsub.fscs@gmail.com - Gmail			2022-04-19 13:58:40	24		-2147483648	0.0
26	20	https://mail.google.com/mail/u/0/				Inbo	
x (12) - unsub.fscs@gmail.com - Gmail			2022-04-19 13:58:40	25		-2147483648	0.0
27	21	https://accounts.google.com/ServiceLogin?service=mail&passive=1209600&				Gmai	
l: Free, Private & Secure Email | Google Works			2022-04-19 13:58:40	26		-2147483648	0.0
28	22	https://mail.google.com/intl/en/mail/help/about.html				Gmai	
l: Free, Private & Secure Email | Google Works			2022-04-19 13:58:40	27		-2147483648	0.0
29	23	https://www.google.com/intl/en/mail/help/about.html				Gmai	
l: Free, Private & Secure Email | Google Works			2022-04-19 13:58:40	28		-2147483648	0.0
30	24	https://www.google.com/gmail/about/				Gmai	
l: Free, Private & Secure Email | Google Works			2022-04-19 13:58:40	29		-1610612736	0.0
31	25	https://accounts.google.com/AccountChooser/signinchooser?service=mail&				Gmai	
l			2022-04-19 13:58:43	0		268435456	0.0

Figure 6.3 — Continuation: the posting-form steps through publication



```
32        26      https://accounts.google.com/AccountChooser?service=mail&continue=https  Gmai
l                                         2022-04-19 13:58:43  31          -2147483648
0.0
33        27      https://accounts.google.com/ServiceLogin?continue=https%3A%2F%2Fmail.g  Gmai
l                                         2022-04-19 13:58:43  32          -1610612736
0.0
34        27      https://accounts.google.com/ServiceLogin?continue=https%3A%2F%2Fmail.g  Gmai
l                                         2022-04-19 13:58:44  0           805306368
0.0
35        28      https://accounts.google.com/signin/v2/identifier?continue=https%3A%2F%  Gmai
l                                         2022-04-19 13:58:44  0           805306368
0.0
36        29      https://accounts.google.com/signin/v2/challenge/pwd?continue=https%3A%  Gmai
l                                         2022-04-19 13:59:26  0           805306368
2967.544
37        30      https://accounts.google.com/CheckCookie?continue=https%3A%2F%2Fmail.go  Inbo
x (12) - unsub.fscs@gmail.com - Gmail     2022-04-19 13:59:29  36          268435456
0.0
38        31      https://mail.google.com/accounts/SetOSID?authuser=0&continue=https%3A%  Inbo
x (12) - unsub.fscs@gmail.com - Gmail     2022-04-19 13:59:29  37          -2147483648
0.0
39        32      https://accounts.youtube.com/accounts/SetSID?ssdc=1&sidt=ALWU2ctdRyPZH  Inbo
x (12) - unsub.fscs@gmail.com - Gmail     2022-04-19 13:59:29  38          -2147483648
0.0
40        19      https://mail.google.com/mail/                                           Inbo
x (12) - unsub.fscs@gmail.com - Gmail     2022-04-19 13:59:29  39          -2147483648
0.0
41        20      https://mail.google.com/mail/u/0/                                       Inbo
x (12) - unsub.fscs@gmail.com - Gmail     2022-04-19 13:59:29  40          -2147483648
0.0
42        33      https://mail.google.com/mail/u/0/#inbox                                 Inbo
x (12) - unsub.fscs@gmail.com - Gmail     2022-04-19 13:59:31  41          1610612736
0.0
43        34      https://mail.google.com/mail/u/0/#inbox/WhctKKXXDrfCqXBGCnHkHGxZnZbkXK  chea
per than Rx supplements - order - unsub.fscs@g  2022-04-19 14:04:12  0           268435456
0.0
```

Figure 6.4 — Continuation: Gmail inbox access and the order thread



```
44        35      https://mail.google.com/mail/u/0/#inbox/WhctKKXXDrfCqXBGCnHkHGxZnZbkXK  chea
per than Rx supplements - order - unsub.fscs@g  2022-04-19 14:04:35  43          1073741824
0.0
45        36      https://mail.google.com/mail/u/0/#inbox/WhctKKXXDrfCqXBGCnHkHGxZnZbkXK  chea
per than Rx supplements - order - unsub.fscs@g  2022-04-19 14:04:46  44          1073741824
0.0
46        34      https://mail.google.com/mail/u/0/#inbox/WhctKKXXDrfCqXBGCnHkHGxZnZbkXK  chea
per than Rx supplements - order - unsub.fscs@g  2022-04-19 14:04:52  45          1610612736
0.0
47        33      https://mail.google.com/mail/u/0/#inbox                                 Inbo
x (12) - unsub.fscs@gmail.com - Gmail     2022-04-19 14:05:59  0           805306368
0.0
48        37      https://mail.google.com/mail/u/0/#inbox/WhctKKXXDrfCqvtRjhGdkMBqVNFnwt  chea
per than Rx supplements -- txid: 517b215691494  2022-04-19 14:56:47  0           268435456
0.0
49        38      https://www.google.com/url?q=https://imgur.com/dTgrkP7&source=gmail&us  Imgu
r: The magic of the Internet              2022-04-19 14:56:52  0           268435456
0.0
50        39      https://imgur.com/dTgrkP7                                               Imgu
r: The magic of the Internet              2022-04-19 14:56:52  49          -1610612736
18926.083
51        40      https://www.google.com/search?q=blockchain+explorer&rlz=1C1ONGR_enUS98  bloc
kchain explorer - Google Search           2022-04-19 14:58:59  0           838860805
0.0
52        40      https://www.google.com/search?q=blockchain+explorer&rlz=1C1ONGR_enUS98  bloc
kchain explorer - Google Search           2022-04-19 14:59:00  0           805306368
1026.231
53        41      https://www.blockchain.com/explorer                                     Bloc
kchain Explorer - Search the Blockchain | BTC   2022-04-19 14:59:01  52          805306368
0.0
54        41      https://www.blockchain.com/explorer                                     Bloc
kchain Explorer - Search the Blockchain | BTC   2022-04-19 14:59:02  0           805306368
18679.186
55        42      https://www.blockchain.com/search?search=517b2156914944339a96137ad8978  Tran
saction: 517b2156914944339a96137ad8978408ea52b  2022-04-19 14:59:06  54          268435456
0.0
```

Figure 6.5 — Continuation: the activity gap and the txid-email thread

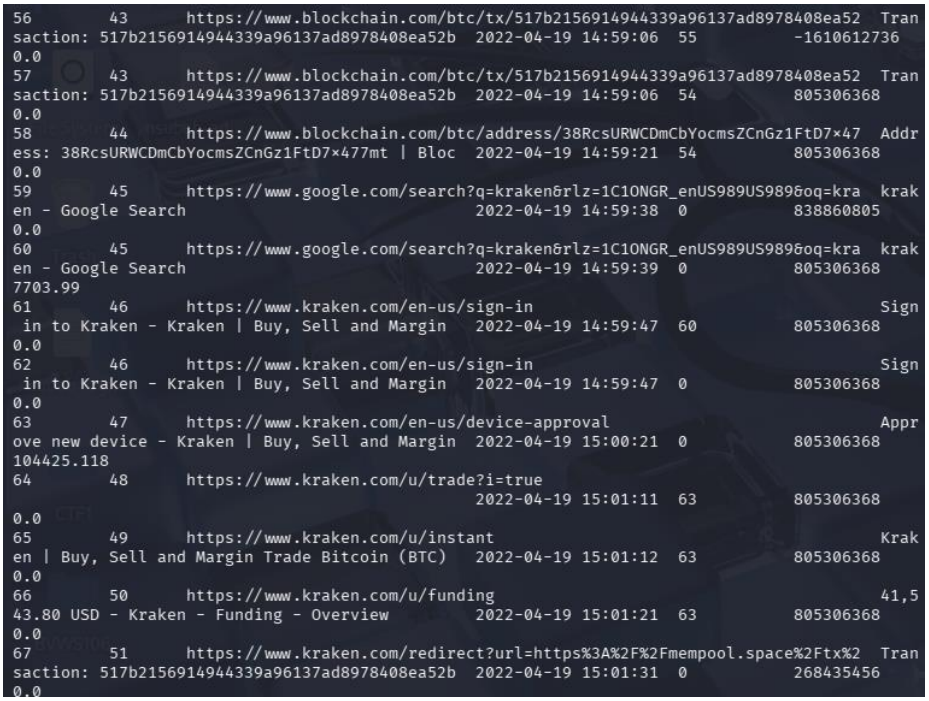


Figure 6.6 — Continuation: Imgur, blockchain.com and Kraken sign-in

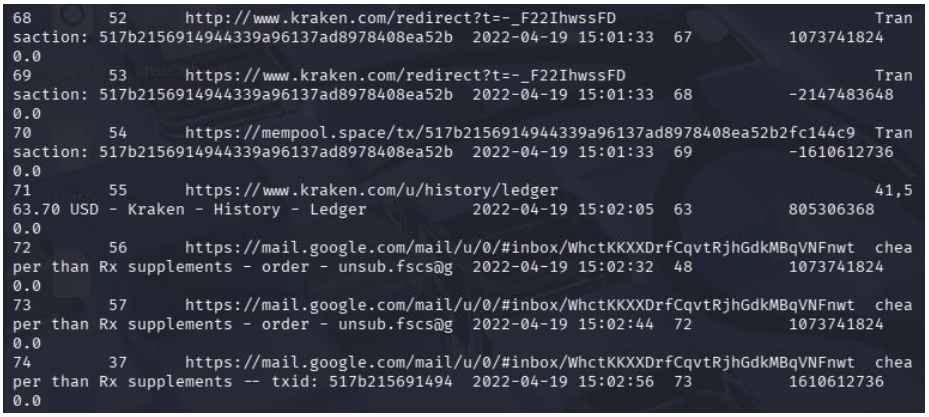


Figure 6.7 — Final rows: Kraken funding/ledger pages and the closing Gmail replies

Stage	UTC window	Visit IDs	Artefact	Observed fact	Interpretation and limit
1. Posting	13:54:18–13:58:31	1–21	Craigslist search, login, 7-step FORM posting, published ad 7473121658	21 visits, 4m13s, posting ID matches between /manage/ and the live ad URL	Strong evidence the profile authored the ad; form content itself not captured
2. Communications (order)	14:04:12–14:05:59	43–47	Gmail "order" thread, 2 reply/compose events	5 visits; thread opened ~5m41s after the ad was published	Thread subject matches the ad title; message body not captured, only the subject line

Stage	UTC window	Visit IDs	Artefact	Observed fact	Interpretation and limit
3. Activity gap	14:05:59–14:56:47	47→48	No recorded browser activity for 50m 48s	Largest idle period in the session	Browser history cannot show what occurred off-browser or on another device during this interval
4. Payment notification	14:56:47–14:56:52	48–49	Gmail txid-email thread; Gmail-sourced redirect to Imgur	Title embeds TXID 517b2156…afdc5 verbatim	Confirms an email referencing the TXID was opened; does not independently confirm the email's full content
5. Imgur proof	14:56:52–14:57:10	50; downloads id=1	imgur.com/dTgrkP7 viewed (18.9s); proof_of_payment.png downloaded (33,844 bytes, complete)	Download tab_url matches the Imgur page visited seconds earlier	Confirms a file was retained from that page; does not independently confirm the image's visual content
6. Blockchain verification	14:59:06–14:59:21	55–58	blockchain.com search, tx permalink (x2), wallet-address lookup	Same TXID looked up on a second, independent explorer	Corroborates the TXID was actively verified, not merely received
7. Exchange / 2nd confirmation	14:59:38–15:02:05	59–71	Kraken sign-in, device approval, funding page (41,543.80 USD), Kraken→mempool.space redirect to the SAME TXID, ledger page (41,563.70 USD)	3rd/4th independent confirmation of the TXID; two balance figures 19.90 USD apart, 6m44s apart	New-device approval suggests first use of this browser profile on this Kraken account; balance-change cause not confirmed by the database

Stage	UTC window	Visit IDs	Artefact	Observed fact	Interpretation and limit
8. Final replies	15:02:32–15:02:56	72–74	Two further reply/compose events on the txid-email thread	Session's final 3 recorded visits	Session ends here; no further activity recorded after 15:02:56 UTC

Stage boundaries above are this report's interpretation, grouping consecutive visits by apparent purpose; they are not a field stored in the database. Every visit ID, timestamp and artefact cited is read directly from the visits/urls/downloads join, and the full 74-row decoded table is reproduced in Appendix C.

Evidence reference: E07a–e.

7. Attribution Assessment (Investigation Question 8)

Investigation question: What conclusion can you defend? This section states a confidence level and sets out limitations and alternative explanations, including the difference between browser-profile activity and human identity, as required by the assignment brief.

7.1 Tier 1 — Direct observations (from the database)

- 74 visits across 57 distinct URLs, spanning 2022-04-19 13:54:18–15:02:56 UTC (4,118 seconds).
- 7 FORM_SUBMIT transitions, all within the Craigslist posting workflow; 4 OMNIBOX transitions (craigslist, gmail, blockchain explorer, kraken); 0 TYPED transitions.
- 6 page titles contain the literal string “Inbox (12) - unsub.fscs@gmail.com - Gmail”, naming the active Gmail account.
- Posting ID 7473121658 is identical between the author-only /manage/ URL (visit 20) and the published ad URL (visit 21).
- Transaction ID 517b2156914944339a96137ad8978408ea52b2fc144c98d3b0b16b21888afdc5 appears, verbatim, in 9 records across 4 platforms (Gmail, blockchain.com, Kraken's redirect, mempool.space).
- downloads.id = 1: proof_of_payment.png (33,844 bytes, complete), sourced from tab_url = https://imgur.com/dTgrkP7, matching the page visited at visit 49–50.
- A 50-minute-48-second gap in recorded browser activity exists between visit 47 (14:05:59 UTC) and visit 48 (14:56:47 UTC).

7.2 Tier 2 — Reasonable inferences

- The browser profile was used, in sequence, to post an advertisement, communicate with an interested party by email, receive a payment notification referencing a Bitcoin transaction, retain an image purporting to show proof of that payment, and verify the same transaction ID on two independent blockchain-explorer services and a cryptocurrency exchange.
- The cross-platform repetition of the same unique transaction ID, within one continuous session and across four independently operated services, reflects a deliberate verification pattern rather than coincidence.

- The device-approval step on Kraken (visit 63) suggests this was the first time this specific browser profile accessed that exchange account.

7.3 Tier 3 — Limitations and alternative explanations

1. Browser-profile activity is not proof of a specific human identity. A Chrome profile, a Gmail session and a Kraken login each identify an account or a profile that was active on this device — not conclusively the particular person physically operating the keyboard at every one of the 74 recorded moments. If the device or profile was shared, left logged in, or accessed by more than one person, the history alone cannot distinguish between them.

2. Page content was not captured. The database stores URLs, titles, timestamps and navigation metadata only — not rendered page content, email bodies, form field values, or chat text. The ad's exact wording, the full text of the Gmail threads, and the visual content of proof_of_payment.png are inferred from titles, filenames and byte counts, not independently verified against the actual page or file content.

3. The 50-minute-48-second gap between visits 47 and 48 is unexplained by this database. Browser history can only show what did not happen in this browser during that window; it cannot show what did happen elsewhere — on another device, in another application, or away from the computer entirely.

4. Redirect chains complicate direct provenance. Several transitions in this session are CLIENT_REDIRECT or SERVER_REDIRECT qualified (for example, the Gmail-to-Imgur hop and the Kraken-to-mempool.space hop) — the visited URL was not always the one the user consciously typed or clicked, and intermediate redirect services could in principle alter a destination without an obvious on-screen cue.

5. URL slugs, page titles and filenames are self-declared by the remote service or the downloading page, not independently verified by this database. A posting title, an email subject line, or a downloaded file's name can be changed by the services involved and are reported here as observed strings, not as verified facts about their content.

6. No corroborating evidence beyond this browser history was examined in this case study. No device identifier, account-registration record, IP address, or physical-world transaction record was available for cross-reference; the conclusions here rest on the History database in isolation.

7. This database is a snapshot in time, hashed and preserved from the moment of acquisition onward. The SHA-256 verification in Section 3 proves the file has not changed since it was acquired for this investigation

— it does not establish what, if anything, occurred to the original device or account activity before or after the captured snapshot.

8. The two Kraken balance figures (41,543.80 USD and 41,563.70 USD) are read from page titles rendered by Kraken's own interface, not from an exported account statement; the 19.90 USD difference between them is noted as an observation and is not attributed to any specific cause by this report.

7.4 Confidence levels

Statement	Confidence
This browser profile was used to post the advertisement, communicate about it, and verify a cryptocurrency transaction in the sequence described in Sections 5–6.	High — directly supported by 74 visit records, 9 cross-platform TXID matches, and the posting-ID match
The activity described is consistent with the case hypothesis of advertising and payment for a controlled item.	Medium — consistent with, but not conclusively proven by, browser metadata alone (page content not captured)
A specific, named human being personally performed every one of these actions.	Low — browser-profile and account activity alone cannot establish who was physically at the keyboard

7.5 Defensible conclusion

The Chrome History database supports, with high confidence, that the examined browser profile was used to publish a Craigslist advertisement, exchange email correspondence referencing an “order” and later a specific Bitcoin transaction ID, retain an image described as proof of payment, and verify that same transaction ID across two blockchain explorers and a cryptocurrency exchange, all within a single 68-minute session on 19 April 2022. This pattern is consistent with the case hypothesis of advertising and receiving payment for a controlled item, and is reported here as a defensible activity pattern with medium confidence. It is not, by itself, conclusive proof that any specific named individual personally carried out every step: browser-profile and logged-in-account activity establish that the device and accounts were used this way, not which human hand was on the keyboard throughout, and the database provides no page-content capture, device-registration record, or independent identity document that would close that final gap. Any further attribution to a named person requires evidence beyond this database — device ownership records, account registration details, or corroborating physical-world evidence — none of which fall within the scope of this case study.

Evidence reference: E08a–d.

8. Appendix A — Evidence Log

Evidence ID	Source record	Finding	Why it matters	SQL / screenshot reference
E01	evidence/History, working/History_working	Source downloaded 2026-10-12; SHA-256/SHA-1/MD5 recorded; working copy byte-identical; original set read-only (chmod 444)	Establishes evidence integrity and chain of custody for all subsequent analysis	Fig. 3.1–3.7 (Section 3)
E02	sqlite_master; PRAGMA table_info(…)	17 tables identified; 4 directly relevant (urls=57, visits=74, downloads=1, downloads_url_chains=1); keyword_search_terms=4	Confirms which tables/fields the findings are drawn from, per Chrome's schema	Fig. 4.1–4.4, 4.8–4.9 (Section 4)
E03	visits.visit_time (all rows)	74 visits span 2022-04-19 13:54:18–15:02:56 UTC (4,118s); local-time offset confirmed as UTC−4 (EDT)	Establishes the UTC timeline base and the conversion method used throughout	Fig. 4.10–4.12 (Section 4)
E04	visits.transition, visits.from_visit (all rows)	9 distinct transition values decoded; 63 LINK / 7 FORM_SUBMIT / 4 OMNIBOX / 0 TYPED; 22 chain-start (from_visit=0) visits	Explains how each page was reached, distinguishing deliberate form entry from passive link-following	Fig. 5.1–5.6 (Section 5.1)
E05	visits/urls, url LIKE '%craigslist%' (21 rows)	7-step posting workflow; posting ID 7473121658 matches between the /manage/ URL and the published ad URL	Ties the browser profile directly to authorship of the suspected advertisement	Fig. 5.7–5.13 (Section 5.2)
E06a	visits 43–47 (Gmail “order” thread)	Order thread opened 5m41s after the ad published; two reply/compose events	Links the posting to buyer communication in the same session	Fig. 5.14 (Section 5.3.1)
E06b	visits 47→48 (50m48s gap)	Largest idle period in the session, between the order thread and the txid thread	Material gap that must be disclosed as a limitation, not glossed over	Fig. 5.16 (Section 5.3.2)
E06c	visits 48, 72–74 (Gmail txid thread)	Email subject embeds TXID 517b2156…afdc5 verbatim	First of 4 independent platforms where this TXID appears	Table in Section 5.3.2
E06d	visits 49–50; downloads id=1	Gmail-sourced redirect to imgur.com/dTgrkP7 (18.9s dwell); proof_of_payment.png (33,844 bytes, complete) downloaded seconds later from the same tab_url	Links the email, the image-hosting page, and the retained file into one evidentiary chain	Fig. 5.17 (Section 5.3.3)
E06e	visits 55–58 (blockchain.com)	Same TXID searched and opened on blockchain.com; wallet 38RcsURWCDmCbYocmsZCnGz1FtD7x477mt looked up	Second independent corroboration of the transaction ID	Fig. 5.18 (Section 5.3.4)

Evidence ID	Source record	Finding	Why it matters	SQL / screenshot reference
E06f	visits 61–71 (Kraken)	Sign-in, new-device approval, funding page (41,543.80 USD), redirect to mempool.space (same TXID), ledger page (41,563.70 USD)	Third/fourth independent confirmation of the TXID; exchange account activity with a dated balance context	Fig. 5.19 (Section 5.3.5)
E07	Full 74-row decoded timeline	7-stage chronological reconstruction separating observed artefact from interpretation	Fulfils the brief's timeline requirement; full table in Appendix C	Fig. 6.1–6.7, Appendix C (Section 6)
E08	Attribution assessment (Section 7)	Tier 1/2/3 separation, 8 stated limitations, confidence table, bounded conclusion	Fulfils the marking guide's requirement for a cautious, evidence-bounded conclusion	Section 7 (this report)

Appendix B — Screenshot Reference

#	Screenshot	Evidence shown
1	1.working_dir.png	Evidence folder structure created before acquisition
2	2.evidence_file.png	wget acquisition of the History file
3	3.hash_evidence.png	SHA-256 / SHA-1 / MD5 of the original file
4	4.working_evidence.png	Working copy creation
5	4.hash_working_evidence.png	SHA-256 of the working copy, confirmed identical
6	5.evidence_protect.png	Original file set read-only (chmod 444)
7	6.file_evidence.png	File-type identification
8	7.Tables.png	Full table listing (17 tables)
9	7.1_schema.png / 7.2_schema.png	PRAGMA table_info() for urls, visits, downloads, downloads_url_chains
10	8.table_counts.png	Row counts across all 5 relevant tables
11	8.visit_preview.png / 8.url_preview1.png / 8.url_preview2.png	Raw table previews of visits and urls
12	8.search_terms.png	keyword_search_terms table (4 rows)
13	8.download_preview.png	downloads table (1 row)
14	9.1–9.7_timeline_utc.png	Full UTC-converted 74-row timeline, 7 screenshots
15	10.date_range.png	Session date-range / span query
16	timezone.png / utc_vs_local.png	Workstation timezone and UTC-vs-local conversion check

#	Screenshot	Evidence shown
17	11.transition_decoded.png / 11.transition_distribution.png	Transition decoding script and base-type distribution
18	12.form_submit.png	FORM_SUBMIT visits isolated
19	13.omnibox_event.png	OMNIBOX visits isolated
20	14.chain_start1.png / 14.chain_start2.png	from_visit = 0 chain-start examples
21	15.craiglist1.png / 15.craiglist2.png / 15.craiglist3.png	Craigslist-domain visit queries
22	16.published_ad.png	Published-ad record detail
23	17.manage_posting.png	Manage-posting record detail
24	18.ad_view.png	Ad-view record detail
25	19.posting_workflow.png / 19.posting2_workflow.png	Full posting-workflow visit sequence
26	20.id_match.png	Posting-ID match between manage URL and published ad
27	21.imgur_proof.png	Gmail-to-Imgur redirect detail
28	22.blockchain_lookups.png	blockchain.com TXID and wallet lookups
29	23.time_gap.png	The 50m48s activity-gap query
30	24.total_visits.png	Kraken session visit sequence
31	25.profile_identity.png	Gmail account-identity title query
32	26.user_input.png	Base-transition count summary (LINK / FORM_SUBMIT / OMNIBOX)
33	evidence_log.png	Contemporaneous evidence_log.md (E01 excerpt)

Appendix C — Completed Timeline Table

Stage	UTC timestamp	Visit / URL ID	Artifact	Observed fact	Interpretation and limit
Posting	2022-04-19 13:54:18	visit 1 / url 1	Google search "craigslist"	OMNIBOX transition, chain start	User-initiated search; not yet on Craigslist
Posting	2022-04-19 13:55:15	visit 15 / url 11	post.craigslist.org … s=edit	FORM_SUBMIT, 177.7s dwell (longest step)	Longest single step in the posting workflow — consistent with composing ad text, not captured
Posting	2022-04-19 13:58:31	visit 21 / url 16	Published ad, posting ID 7473121658	LINK, matches /manage/ ID from visit 20	Confirms authorship link to the profile that logged in at visit 6
Communications	2022-04-19 14:04:12	visit 43 / url 34	Gmail "order" thread opened	LINK, chain start, ~5m41s after ad published	Subject matches ad title; body content not captured
Activity gap	2022-04-19 14:05:59 → 14:56:47	visit 47 → 48	No recorded visits for 50m48s	Largest gap in the session	Cannot be explained from this database alone
Payment notice	2022-04-19 14:56:47	visit 48 / url 37	Gmail txid-email thread opened	Title embeds TXID verbatim	Confirms an email referencing the TXID was opened
Imgur proof	2022-04-19 14:56:52	visit 49–50 / url 38–39	Redirect to imgur.com/dTgrkP7	18.9s dwell on the Imgur page	Matches the download's source tab_url
Download	2022-04-19 14:56:58–14:57:10	downloads id=1	proof_of_payment.png saved	33,844 / 33,844 bytes — complete download	File content (image) itself not independently verified by this report

Stage	UTC timestamp	Visit / URL ID	Artifact	Observed fact	Interpretation and limit
Blockchain check	2022-04-19 14:59:06	visit 55–57 / url 42–43	blockchain.com TXID search and permalink	Same TXID as the Gmail thread	Second independent platform confirming the TXID
Exchange	2022-04-19 14:59:47–15:02:05	visit 61–71	Kraken sign-in, device approval, funding/ledger pages	Two balance figures 19.90 USD apart, 6m44s apart	Balance-change cause not confirmed by the database
2nd TXID confirm	2022-04-19 15:01:31–33	visit 67–70 / url 51–54	Kraken redirect → mempool.space, same TXID	Third/fourth independent confirmation	Redirect-qualified transition; destination not directly typed by the user
Final replies	2022-04-19 15:02:32–15:02:56	visit 72–74	Two further replies on the txid-email thread	Session's final recorded activity	No activity recorded after 15:02:56 UTC in this database

Appendix D — Completed Evidence Log (Template)

Evidence ID	Source record	Finding	Why it matters	SQL / screenshot reference
E01	evidence/History	Acquired, hashed (SHA-256/1, MD5), working copy verified identical, original set read-only	Chain of custody for the whole investigation	Figures 3.1–3.7
E02	sqlite_master / PRAGMA	17 tables; 4 directly relevant; row counts confirmed	Scopes which tables/fields support the findings	Figures 4.1–4.4
E03	visits.visit_time	74 visits, 2022-04-19 13:54:18–15:02:56 UTC, EDT offset confirmed	Establishes the timeline's time base	Figures 4.10–4.12
E04	visits.transition / from_visit	63 LINK, 7 FORM_SUBMIT, 4 OMNIBOX, 0 TYPED; 22 chain starts	Explains navigation method for every finding	Figures 5.1–5.6
E05	Craigslist-domain visits (21)	Posting-ID match, manage URL ↔ published ad	Links profile to ad authorship	Figures 5.7–5.13
E06	Gmail / Imgur / downloads / blockchain.com / Kraken (visits 43–74, downloads id=1)	TXID appears 9 times across 4 platforms; proof-of-payment file retained	Core communications/payment correlation	Figures 5.14–5.19